

BOND VALUATION AND TRADING

(TYPES, RISK TYPES, MEASURING INTEREST RATE RISK, EMBEDDED OPTIONS,
TRADING CONVENTIONS)

Source:

- Investments Part IV., Risk Management and Financial Institutions 4. ed. Ch. 9.
- *Investments*, BKM, 12th Ed., Ch. 2.2
- Hull book, Ch. 4
- *Principles of Corporate Finance*, Brealey, Myers, Allen, Edmans (14th Ed.): Ch 3

The bond market

The bond market is composed of longer term borrowing, and shorter term debt instruments that trade in the money market. This market is also called the fixed income capital market because most of the instruments promise a fixed stream of income or a stream of income that is determined by a formula.

A **bond** is a security that is issued in connection with a borrowing arrangement. The borrower issues (i.e. sells) a bond to the lender for some amount of cash. The issuer agrees to make specified payments to the bondholder on specified dates. This is the coupon payment. When the bond matures the issuer repays the debt by paying the bond's par value (also called face value). The coupon rate of the bond determines the interest payment.

Zero coupon bonds make no coupon payment. These bonds are issued below par value, and the bondholder receives the par value upon maturity.

BOND TYPES

Government bonds: *Treasury Bills, Notes and Bonds.* T-Bills are issued w/ maturities up to 1 year, sold at discount. T. Notes are issued with maturities between 1 to 10 years, while T. Bonds are issued with maturities between 10 to 30 years. Both may be purchased directly from the Treasury and are denominated in USD making semiannual coupon payments. Bid and ask prices are quoted as a percentage of par value.

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Treasury Inflation-Protected Securities (TIPS): bonds that are linked to an index of the cost of living to provide an effective way to hedge inflation risk.

Municipal bonds: are issued by states and local governments.

Corporate Bonds: Corporations borrow money by issuing bonds, just like governments, directly from the public. Most bonds still trade on a traditional over-the-counter market. Companies may have dozens of different bonds outstanding which may trade only a few times a year, so they are rather illiquid. Based on the credit rating we can differentiate between investment-grade and junk-bond.

If a firm borrows abroad, then we talk about eurobonds. These are bonds denominated in a currency other than that of the country in which it is issued. A Yankee bond is a dollar-denominated bond sold in the US by a non-US issuer. Samurai bonds are yen-denominated bonds sold in Japan by non-Japanese issuers.

Floating-Rate Bonds: make interest payments tied to some measure of current market rates. (e.g. rate adjusted annually to the current T-bill rate plus 2%)

Zero-coupon bonds: they pay no coupon but are sold at a discount and pay the par value at maturity.

Mortgage and Asset-Backed securities: it is either an ownership claim in a pool of mortgages or an obligation that is secured by such a pool.

Other bond types: International Bonds, Inverse Floaters, Asset-Backed Bonds, Catastrophe Bonds, Indexed Bonds.

EMBEDDED OPTIONS

An embedded option is a function that allows holders or issuers of financial securities to take specified action against one another in the future.

Callable Bonds: Callable means that it allows the issuer to repurchase the bond at a specified call price before the maturity date. The bondholder has essentially sold a call option to the

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company that issued the bond. It is favorable for the issuer company because if interest rates decrease, they can call the higher-cost bonds and can issue new bonds at lower cost. This is why investors are compensated with higher coupons than for non-callable bonds. They typically come with a period of *call protection*, an initial time during which the bonds are not callable (deferred callable bonds).

Puttable Bonds: Gives the option to the bondholder to ask for repayment of the par value. Owners of puttable bonds have essentially purchased a put option built into the bond. Ex. if the coupon on the bond is significantly lower than rates on the market, the holder would ask for repayment and invest the money in newly issued securities with higher rates. Puttable bonds offer lower yields compared to non-puttable bonds.

Convertibles

- **Convertible Bonds:** give bondholders an option to exchange each bond for a specified number of shares of common stock of the firm. Conversion ratio is the number of shares for which each bond may be exchanged. This bond combines fixed income with potential equity upside. Typically exercised if the issuer's stock price rises significantly. Convertible bonds generally offer lower yields than regular corporate bonds.
- **Convertible Preferred Stock:** preferred shares that include an option for the holder to convert them into a fixed number of common shares after a predetermined date. Preferred stock has a higher claim on assets and earnings than common stock. Preferred shares typically pay steady dividends, while common stock pays dividends only if and when they are approved by the board of directors based on the company's recent financial performance.

Bond Pricing

To value a security, we discount its expected CFs by the appropriate discount rate.

$$\text{Bond value} = PV(\text{coupons}) + PV(\text{par value})$$

$$\text{Bond value} = \sum_{t=1}^T \frac{\text{Coupon}}{(1+r)^t} + \frac{\text{Par value}}{(1+r)^T}$$

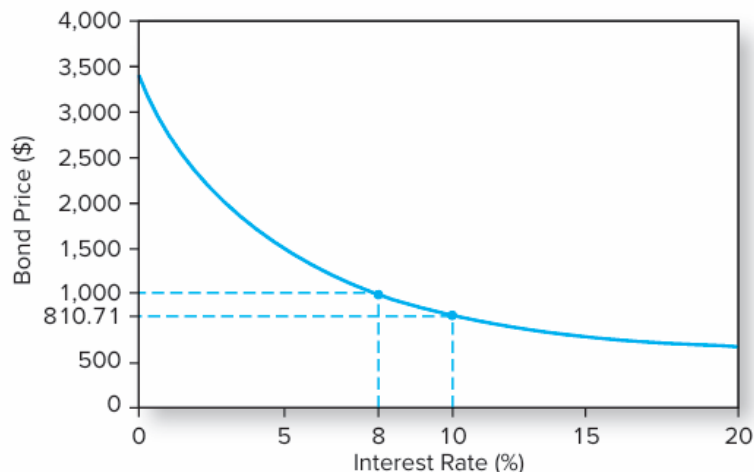
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If we use a **single discount rate**: the first term is the **PV of an annuity**, the second term is the PV of a *single term*, the face value. The formula for the whole price is:

$$\text{Price} = \text{Coupon} \times \frac{1}{r} \left[1 - \frac{1}{(1+r)^T} \right] + \text{Par value} \times \frac{1}{(1+r)^T}$$

When a **bond is issued** the **coupon rate** is chosen to be approximately the **market yield**. After the bond is issued it can be **traded on secondary markets** where **its price fluctuates inversely compared to the market interest rates**: an **increase in the int. rate** results in a **price decline**, a **drop in the interest rate** results in a **price increase**. The **price decline from an increasing interest rate** is always **smaller than** the **price gain resulting from the same decrease in int. rate** (chart below). **Interest rate fluctuation** is the main source of risk in the fixed-income market measured by sensitivity. Generally, **the longer the maturity (and duration) of the bond the higher its sensitivity** bc the longer the money is tied up, the **greater the potential loss if interest rates increase**, so the greater the drop in the bond price.



Sometimes traders use the **same DF for each CF**, but a more accurate approach would be to use **different zero rates for each CF** (zero rates can be calculated using **bootstrapping**). A coupon bond can be valued using **treasury stripping**. Stripped Treasuries are zero-coupon bonds created by selling each coupon or principal payment from a whole Treasury bond as a separate CF. So, if each CF can be sold off separately, then the value of the whole bond should equal the total value of its CFs bought piece by piece in the STRIPS market. If the two are not equal, there would be an arbitrage opportunity:

- If the bond were selling for less than the sum of the separate CFs, then someone could buy the bond for less and sell the stripped CFs, thus making a profit (*bond stripping*)

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- If the bond were selling for more than the sum of the separate CFs, then someone could buy the individual CFs in the STRIPS market and sell the bond for more, thus making a profit (*bond reconstitution*)

TRADING CONVENTIONS

Bonds are issued by entities such as governments, municipalities, or corporations to raise funds. The issuance is usually facilitated by underwriters (investment banks) who sell the bonds to institutional investors (pension funds, insurance companies etc.) in large quantities. After issuance, bonds can be bought and sold in the secondary market, which operates either *over-the-counter (OTC)* or on *exchanges* (for some corporate and government bonds). A network of bond dealers quote the prices at which they are ready to buy/sell. If one wishes to trade a bond, she contacts her broker, which will contact the dealer, and the trade is done.

Accrued interest and Quoted Bond Prices: Bonds are generally quoted as a percentage of their face value (par value). Usually, the bond prices quoted are not the actual prices that investors pay bc they do not include the interest that accrues between coupon payment dates (*flat price / clean price*). If a bond is purchased between coupon payment dates the buyer must pay the seller for accrued interest (*invoice price / dirty price*):

$$\text{Accrued interest} = \frac{\text{Annual coupon payment}}{2} \times \frac{\text{Days since last coupon payment}}{\text{Days separating coupon payments}}$$

Day counts: defines the way in which interest occurs over time. It is defined as X/Y , where X shows how the number of days between the two coupon dates are calculated, and Y shows how the total number of days in the reference period is measured.

$$\frac{\text{Number of days between dates}}{\text{Number of days in reference period}} \times \text{Interest earned in reference period}$$

- *Actual/actual* (in period): used for US Treasuries and government bonds
- *30/360*: corporate and municipal bonds
- *Actual/360*: US money market instruments
- *Actual/365*: UK government bonds

Coupon payment: defines the frequency of coupon payments

- *Annual/Quarterly/Semi-Annual Payments*: Most bonds pay interest semi-annually, though some pay annually or quarterly.

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- **Fixed vs. Floating Rate:** Fixed-rate bonds have a set interest payment over the life of the bond, while floating-rate notes (FRNs) adjust periodically based on a reference rate.

Bond Yields

A bond's yield is the single discount rate that, when applied to all CFs, gives a bond price equal to its market price. When traded, bonds mostly do not sell for par value, but they will mature to par. Yield to Maturity is a measure of the total rate of return. **Yield to Maturity** is defined as the interest rate that makes the present value of a bond's payments equal to its price. It is a measure of *average return* to an investor who purchases the bond for the ask price and holds it until its maturity date. Yield to maturity equals the *rate of return realized* over the life of the bond if all coupons are reinvested at the same rate. It is also an IRR to a bond investment.

Current Yield: Bond's annual coupon payment divided by its price.

General Rules: For premium bonds (bonds selling above par value): Coupon rate > current yield > YTM. For discounted bonds (bonds selling below par value): Coupon rate < current yield < YTM.

Par yield: the par yield for a certain bond maturity is the *rate* that causes the bond price to equal its par value.

Forward interest rates: rates of interest implied by the current zero rates for periods of time in the future. For ex. a year 2 forward rate is the int. rate implied by the zero rates for the period of time between the end of the first year and the end of the second year. If 1 year zero rate is 3% and 2 year zero rate is 4% then the year 2 forward rate is that, when combined with 3% per annum for year 1, gives 4% overall for the 2 years. So it is 5% in this case.

Yield to Call: if interest rates fall, the bond price, which equals the present value of the promised payments, can rise substantially. If there is a bond callable at 110%, when interest rates fall, the PV of the bond (par value 1000) increases, but the call provision allows the issuer to repurchase the bond at the call price. If the call price is less than the PV of the bond, the issuer may call the bond back from the bondholder, therefore theoretically the PV of the bond is unlikely to go above 110% (1100).

When interest rates change, bond investors are subject of two offsetting sources:

- When rates rise, bond prices fall, i.e. portfolio value decreases.

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- When rates rise, reinvested coupon income will compound more rapidly at those higher rates.

Yield to Maturity vs. Holding-Period Return: When rates are constant the two will be equal, but when market rates fluctuate the price of the bond will also change, thus the holding period return can differ from the initial yield to maturity of the bond. YTM depends only on the bond's coupon, current price and par value at maturity, and all of these are observable today. HPR is the *rate of return over a particular investment period* and depends on the market price of the bond at the end of that holding period, which price is not known today. As we should incorporate in the holding-period return the *received coupon payments and the price appreciation/depreciation*.

Consider a 30-year bond paying an annual coupon of \$80 and selling at par value of \$1,000. The bond's initial yield to maturity is 8%. If the yield remains at 8% over the year, the bond price will remain at par, so the holding-period return also will be 8%. But if the yield falls below 8%, the bond price will increase. Suppose the yield falls and the price increases to \$1,050. Then the holding-period return is greater than 8%:

$$\text{Holding-period return} = \frac{\$80 + (\$1,050 - \$1,000)}{\$1,000} = .13, \text{ or } 13\%$$

The Term Structure of Interest Rates

Term structure: it is the *relationship between short- and long-term (spot) interest rates*. When the term structure of spot rates is upward (downward) sloping, the yields (YTM) increase (decrease) with maturity.

Yield curve: a *plot of YTM as a function of TTM for bond of the same credit quality*. It is the graphical representation of the term structure. Typically, the curve is upward-sloping, but it can be hump-shaped (first arising then falling), inverted (downward sloping) or even flat.

- *Pure yield curve* refers to the curve for stripped or zero-coupon bonds.
- *On-the-run yield curve* refers to the plot of yield as a function of recently issued coupon bonds selling at or near par value.

Yield curve under certainty

In a certain word (with no risk) all bonds (in fact, all securities) must offer the same returns. In this word, *an upward-sloping yield curve is evidence that short-term rates are going to be higher next year than they are now*.

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$$(1 + y_2)^2 = (1 + r_1) \times (1 + r_2)$$

$$1 + y_2 = [(1 + r_1) \times (1 + r_2)]^{1/2}$$

- **Spot rate:** YTM on zero-coupon bonds today (the rate that prevails today for a time period corresponding to the zero's maturity).
- **(Future) short rate:** for a given interval refers to the interest rate for that interval available at different points in time. So, the 2-year spot rate today is an average of the short rate of today and next year. In a certain world two different investment strategies with common terminal dates must provide equal rates of return.

$$(1 + y_2)^2 = (1 + r_1) \times (1 + r_2)$$

$$1 + y_2 = [(1 + r_1) \times (1 + r_2)]^{1/2}$$

When next year's short rate, r_2 , is greater than this year's short rate, r_1 , the average of the two rates is higher than today's rate, so $y_2 > r_1$ and the yield curve slopes upward. If next year's short rate were less than r_1 , the yield curve would slope downward. Thus, the yield curve, to some extent, reflects the market's assessments of coming interest rates.

We can infer future short rate from the yield curve of zero-coupon bonds.

$$(1 + y_n)^n = (1 + y_{n-1})^{n-1} \times (1 + r_n) \quad (15.2)$$

where n denotes the period in question and y_n is the yield to maturity of a zero-coupon bond with an n -period maturity. Given the observed yield curve, we can solve Equation 15.2 for the short rate in the last period:

$$(1 + r_n) = \frac{(1 + y_n)^n}{(1 + y_{n-1})^{n-1}} \quad (15.3)$$

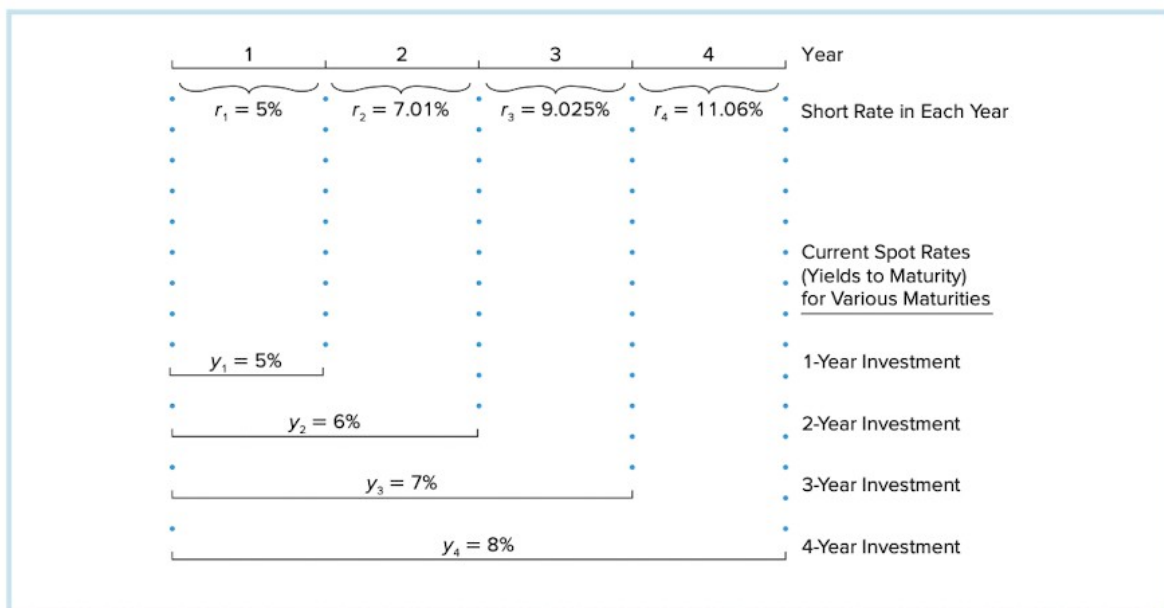


Figure 15.3 Short rates versus spot rates

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Uncertainty: but instead of referring to it as “future” short rate, we call it *forward interest rate* to account for uncertainty, denoted by f . The forward rate is defined as the “break-even” interest rate that equates the return on an n -period zero-coupon bond to that of an $(n-1)$ -period zero-coupon bond rolled over into a 1-year bond in year n . In case of certainty, future short rates equal forward interest rates, so the actual total returns on the two n -year strategies will be equal. *When there is risk (bond prices reflect a risk premium), the interest rate that will prevail in the future need not equal the forward rate.* Actually, the forward rate exceeds the expected short rate because the lower the expectation of r_2 , the greater the anticipated return (HPR) on the long-term bond. If most individuals are short-term investors, then $f_2 > E(r_2)$. The forward rate will embody a premium (*liquidity premium*) compared to the expected future short-interest rate, to compensate short-term investors for the uncertainty about the price at which they will be able to sell their long-term bonds at the end of the year. Contrary, if all investors are long-term investors, then $f_2 < E(r_2)$ because no one would be willing to hold short-term bonds unless rolling over those bonds offered a reward for bearing interest rate risk.

Theories of the Term Structure

Expectations Hypothesis

- This theory suggests that the shape of the yield curve reflects market *expectations about future short-term interest rates*. The *only* reason for an upward-sloping (downward-sloping) term structure is that investors expect short-term interest rates to rise (fall).
- It states that in well-functioning bond markets *investment in a series of short-maturity bonds must offer the same expected return as an investment in a single long-maturity bond*. For example, if the investor is considering a 5-year bond, the yield on that bond should reflect the average of the expected 1-year rates for each of the next five years. Thus, investors are indifferent to holding a long-term bond versus a series of short-term bonds.
- Equivalently, it states that *the forward rate equals the market consensus expectation of the future short interest rate; $f_2 = E(r_2)$ and liquidity premiums are zero*. Therefore, we can use forward rates derived from yield curve to infer market expectations about future short rates. An upward-sloping yield curve would be evidence that investors expect increases in interest rates.

Market Segmentation Theory

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- There is no need for a relationship between short-term and long-term interest rates. The *short-term interest rates are determined by supply and demand in the short-term bond market*, and *long-term interest rates are determined by supply and demand in the long-term bond market* (the case of pension funds or life insurance companies: they buy long-term bonds and hold them until maturity, because they are better off with a fixed inflow stream that covers their fixed cash-outflows).

Liquidity Preference Theory

- Expectations are not the only factor that drives the term structure, *risk* is also important. *Investors* (except pension funds or life insurance companies) *are prepared to hold long bonds only if they offer a compensation of higher prospective returns against interest rate movements*. In this case, the term structure will be upward sloping even if interest rates are expected to remain the same.
- It assumes that *investors* prefer short term bonds because they are more liquid, while *borrowers* prefer to borrow at fixed rates for longer periods of time. To encourage investors to hold longer-term bonds, issuers offer a liquidity premium.
- As a result, long-term bond yields are generally higher than short-term yields, even if future short-term rates are expected to remain stable. This leads to a typically upward-sloping yield curve.
- Short-term investors are unwilling to hold long-term bonds unless the forward rate exceeds the expected short interest rate: $f_2 > E(r_2)$, so there is a liquidity premium.

RISK TYPES

Interest Rate Risk

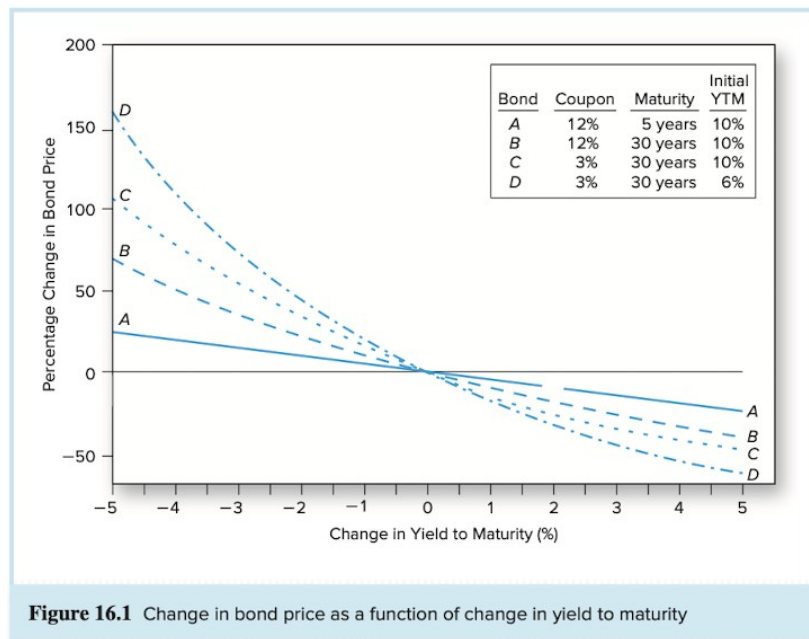
There are 6 key concepts of sensitivity:

1. Bond prices and yields move in opposite directions.
2. An increase in a bond's YTM results in a smaller price change than a decrease in yield of equal magnitude.
3. Prices of long-term bonds tend to be more sensitive to interest rate changes than prices of short-term bonds.
4. The sensitivity of bond prices to changes in yields increases at a decreasing rate as maturity increases (*can be measured by second derivative*). Interest rate risk is less than proportional to bond maturity.

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5. Interest rate risk is inversely related to the bond's coupon rate. Prices of low-coupon bonds are more sensitive (higher duration) to changes in interest rates than prices of high-coupon bonds.
6. The sensitivity of a bond's price to a change in its yield is inversely related to the yield to maturity at which the bond currently is selling. A bond with higher YTM is less sensitive to interest rate changes (higher YTM diminishes the value of CFs in far future).



Duration / Macaulay's duration: it measures *how long the holder of the bonds has to wait before receiving the PV of the CF payments*. It is the average maturity of the bond's promised CFs. It is a weighted average of the times to each coupon or principal payment, with weights related to the "importance" of that payment to the value of the bond. All discounting is done at the bond yield rate (not diff. zero rates for each CF).

$$w_t = \frac{CF_t / (1+y)^t}{\text{Bond price}} \quad D = \sum_{t=1}^T t \times w_t$$

It can be also measured with the following formula:

$$D = -\frac{1}{B} \frac{\Delta B}{\Delta y}$$

so that

$$\Delta B = -DB\Delta y$$

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When the yield of the bond is measured with cont. comp. then the two should give the same value.

A zero-coupon bond that lasts n years has a duration of n years. For a coupon-bearing bond it is less than n bc the holder gets some of the cash payments prior to year n . *The higher the duration the more sensitive the bond is to interest rate changes.* To measure how bond prices change when interest rates change, we need to use **modified duration**:

$$D^* = \frac{D}{1 + y/m} \quad \Delta B = -BD^* \Delta y$$

Modified duration *measures the percentage change in bond price for 1% (percentage point) change in yield*. However, it is only approximately valid for small changes in the bond's yield.

The duration of a **bond portfolio** can be defined as a weighted avg of the durations of the individual bonds. In this case, there is an implicit assumption that the yields of all bonds will change by approx. the same amount. When bonds have widely differing maturities, this happens only when there is a parallel shift in the zero-coupon yield curve.

Duration is important because:

- Shows the effective average maturity of a portfolio
- A measure of the interest rate sensitivity of a portfolio
- Is an essential tool in *immunizing portfolios* from interest rate risk

Rules of Duration determinants:

1. Duration of a zero-coupon bond equals to its time to maturity.
2. Holding maturity constant, a bond's duration is lower when the coupon rate is higher.
3. Holding coupon rate constant, a bond's duration generally increases with its time to maturity.
4. Holding other factors constant, duration of a coupon bond is higher when the bond's YTM is lower.
5. Duration of a perpetuity: $(1+y)/y$ where y = yield

Immunization:

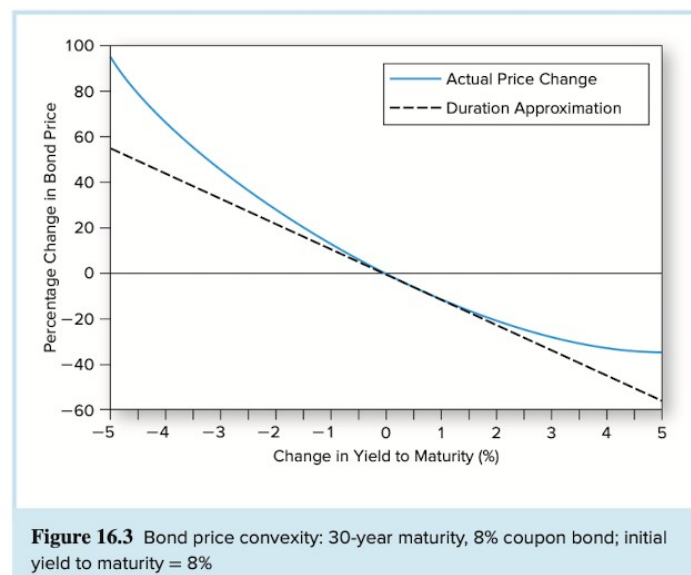
It is a strategy used by investors to *shield their overall financial exposure to interest rate risk*. For ex. banks may want to match the interest rate exposure of their assets and liabilities so that

the value of assets will track the value of liabilities if rates fluctuate. Usually, their liabilities are short-term deposits with low duration, while assets have longer duration. Thus, when interest rates increase, assets fall more in value than liabilities.

One possible solution for an investor to immunization is to buy zero-coupon bonds with face value equal to the projected cash outflows (cash flow matching). Cash flow matching on a multiperiod basis is referred to as a *dedication strategy*.

Convexity:

Duration rule is only an approximation. It applies only to *small changes* in yields. The *relationship between bond prices and yields is not linear*. So, bond price change is not directly proportional to the bond's yield. In this case, duration approximation always understates the value of the bond. Instead, there is a *convex relationship* between the two (bond price and yield changes).



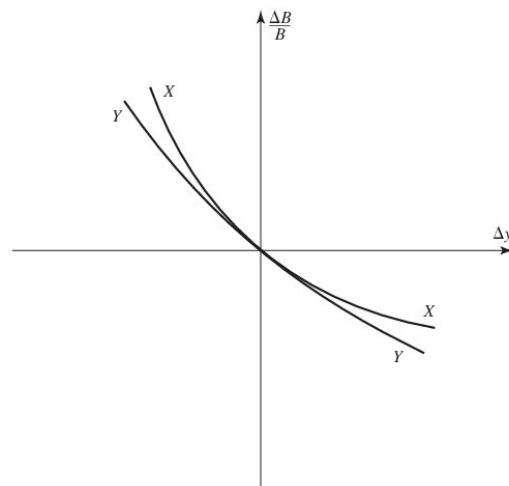
In bond valuation, **convexity** is a measure of how the duration of a bond changes as interest rates change. Differently said, it is the rate of change of the slope of the price-yield curve, expressed as a fraction of the bond price. It provides insight into the *relationship between the percentage change in a bond's price and change in yield*, which affects how sensitive the bond is to changes in interest rates. Convexity is an important tool for assessing the interest rate risk of a bond.

$$C = \frac{1}{B} \frac{d^2 B}{dy^2} = \frac{\sum_{i=1}^n c_i t_i^2 e^{-y t_i}}{B}$$

$$\frac{\Delta B}{B} = -D\Delta y + \frac{1}{2}C(\Delta y)^2$$

Portfolio X and Y change in value by the same percentage for small yield changes but behave differently for large yield changes. X is more convex than Y, so it enjoys greater price increases and smaller price decreases when interest rates fluctuate by larger amounts.

Figure 4.2 Two bond portfolios with the same duration.



Inflation risk

Someone saving for retirement can decide between buying a long-term bond or investing in a succession of short-term bonds. In case of a long-term bond it is exactly known how much money it will give during the years, but not how much that money will be worth. Uncertainty about inflation may make it risky to invest in longer-term bonds. Investment in short-term bonds and rolling them over could be a better decision since future interest rates will adapt to inflation. Another solution can be the purchase of TIPS.

Credit (Default) Risk

Uncertainty about the CFs of a bond is minimal as long as the issuer is sufficiently creditworthy. Bond income stream is not guaranteed unless the investor can be sure the issuer will not default on the obligation i.e. there is default risk, which is usually called *credit risk*, and it is measured by rating agencies like Moody's, S&P Corp and Fitch. They analyze the company's performance (coverage ratios, leverage, liquidity ratios, profitability ratios and CF-to-debt ratios) and assign credit ratings. There are two major rating categories:

- BBB and above: *investment-grade bonds*

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- below BBB: *speculative-grade bonds* or junk bonds (high-yield bonds)

Because corporate bonds are subject to default risk, there is a *difference between promised YTM and expected yield*. Thus, promised YTM is the maximum possible YTM of the bond. When a bond becomes more subject to default risk, its price will fall, and the promised YTM will rise. Similarly, the yield spread of the bond (stated YTM – T bond) will rise. However, the expected YTM will be far less. To compensate for the possibility of default, corporate bonds must offer a *default premium (credit spread)*. This is the difference between the promised yield and the yield on a riskless, otherwise-identical government bond.

Credit Default Swaps is an insurance against the default of a bond, allowing lenders to buy protection against default risk.

Liquidity Risk

The risk that a bond cannot be sold quickly at its fair market value.

Call (Prepayment) Risk

The risk that a bond issuer will redeem a bond before its maturity, typically when interest rates fall. In this case investors must reinvest the returned principal at lower interest rates.

Currency Risk

The risk that exchange rate fluctuations will affect the value of a bond denominated in a foreign currency.

Liquidity risk

Interest rate risk management is a key activity for a financial institution, which is incorporated in the *management of net interest income* (that is the excess of interest received over interest paid). The role of the asset-liability management function within the institution is to ensure that the *net interest margin*, which is net interest income divided by income-producing assets, remains roughly constant over time. It can minimize risk by *ensuring that the maturity of assets on which interest is earned is matched to the maturities of liabilities on which interest is paid*. If there is a mismatch of assets and liabilities in terms of maturities, it can lead to liquidity problems. A bank that funds long-term loans with short-term deposits has to replace maturing

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deposits with new deposits on a regular basis. If depositors lose confidence in the bank, it might find it difficult to do this (bank runs).